

Online Influencer Product Recommendation System

Module 3: Application of the model and the interactive tool

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Project Summary

Background: Social media and e-commerce integration have made influencers central to online sales. However, choosing the right products to promote remains challenging, requiring a balance between market trends, personal brand, and audience preferences.

Project Goal: Develop a data-driven product recommendation system for social media influencers to enhance product promotion on E-Commerce platforms.

The system aims:

- 1) To boost sales and commission earnings
- 2) To improve brand partnerships through more effective product placements.



Who are we designing for?

Target Audience — Instagram Influencers

- Follower count: 10K–1M+
- Active in product promotion / brand partnerships
- Want efficiency in selecting products
- Non-technical, visual learners



Our tool empowers influencers to **make smarter, faster product promotion decisions** aligned with their **brand identity** and **follower preferences**.

Audience Demand

Influencer Challenges

- Too many products, hard to choose
- Uncertain about what fans will like
- No time to browse E-commerce product catalogs
- Doesn't know how to track performance

What our tool achieves

- Extract a list of top recommended products with customized varieties
- Matching with similar influencer's promotion & past engagement
- Fast recommendations via our interactive tool
- Provide predicted likes and recommendation scores through our model

Recall – Model Functionality

- Input : Influencer ID/Name
- Stage1 : Recall
 - Collaborative Filtering # TF-IDF + Cosine Similarity + KNN
 - History-based recall # Influencer past promotions
 - Popularity-based recall # Global promotion count filter
- Stage2 : Fine Ranking
 - Relevance Scoring = $\alpha \times \text{Predicted Likes} + \beta \times \text{Similarity} + \gamma \times \text{Product Fit}$
 - Engagement prediction # LightGBM
- Stage 3: Re-ranking
 - Embedding Matching with Amazon Products
 - SBERT Embeddings # Used for influencer caption & product vectorization
 - FAISS similarity search
- Output: Top-N Product Recommendations + Best Posting Time

Interactive Tool Illustration – Interface



Influencer Product Recommendation System

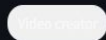


Influencer Profile



shanedawson

@shanedawson



7,300,391.0

Followers

204,349,743

Total Likes

1,053,349.2

Avg Likes/Post

2841.80%

Engagement Rate

Latest Post:



Recommended Products

Recommendation 1: movie



Recommendation 2: movie



Recommendation 3: movie



Select Influencer

shanedawson (ID: 171146)



Score Weights

Predicted Engagement

0.55

0.00

1.00

Similar Influencer Recommendation

0.80

0.00

1.00

Influencer Product Fit

0.25

0.00

1.00

Product Categories

Filter by Categories (Optional)

Choose an option



Choose from Instagram Hot Products

Only Recommend Hot Products?

No



Amazon Product Recommendations

Number of Amazon products to show

3

1

10

Select Influencer

shanedawson (ID: 171146) ▼

Score Weights

Predicted Engagement

0.00 0.55 1.00

Similar Influencer Recommendation

0.00 0.80 1.00

Influencer Product Fit

0.00 0.25 1.00

Product Categories

Filter by Categories (Optional)

Choose an option ▼

Choose from Instagram Hot Products

Only Recommend Hot Products?

No ▼

Amazon Product Recommendations

Number of Amazon products to show

1 3 10

Tool Elements — Input section

Influencer Profile Selection

Select Influencer

ha ▼

hayesgrier (ID: 928481404)

hasbulla.hushetskiy (ID: 44611043...)

misha (ID: 36298923)

ashanti (ID: 7950648)

cthagod (ID: 11884916)

tonyhawk (ID: 1132845)

ihansika (ID: 313548305)

londongha (ID: 4701510)

Weight adjustments corresponding to
 $\alpha \times \text{Predicted Likes} + \beta \times \text{Similarity} + \gamma \times \text{Product Fit}$

Score Weights

Predicted Engagement

0.00 0.55 1.00

Similar Influencer Recommendation

0.00 0.80 1.00

Influencer Product Fit

0.00 0.25 1.00

Output preference (choose from 33 categories of all Amazon products)

Product Categories

Filter by Categories (Optional)

Choose an option ▼

Movies_and_TV

Clothing_Shoes_and_Jewelry

Others

Grocery_and_Gourmet_Food

Health_and_Personal_Care

Digital_Music

na

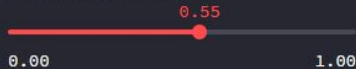
Software

Select Influencer

shanedawson (ID: 171146)

Score Weights

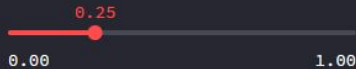
Predicted Engagement



Similar Influencer Recommendation



Influencer Product Fit



Product Categories

Filter by Categories (Optional)

Choose an option

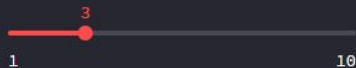
Choose from Instagram Hot Products

Only Recommend Hot Products?

No

Amazon Product Recommendations

Number of Amazon products to show



Tool Elements — Input section cont.

Choose from Instagram Hot Products

Only Recommend Hot Products?

Yes

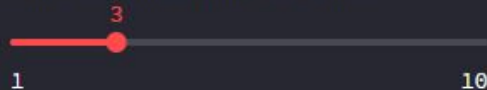
No

Yes

Whether output products are limited to the Hot Product List

Amazon Product Recommendations

Number of Amazon products to show



Number of output products in the recommendation list

Tool Elements — Output section



Recommended Products ↔

Recommendation 1: movie

Category: Movies_and_TV

Brand: na

Product Info: movies_and_tv movie na

Predicted Likes(k): 1,465

Predict likes by day of week

Comprehensive Score: 0.70

Recommendation Score: 0.15

Product Fit: 0.29

Engagement Change: 39.10%

Recommendation 2: movie

Category: Movies_and_TV

Brand: 20th century studios

Product Info: movies_and_tv movie 20th century studios

Predicted Likes(k): 1,448

Predict likes by day of week

Comprehensive Score: 0.68

Recommendation Score: 0.02

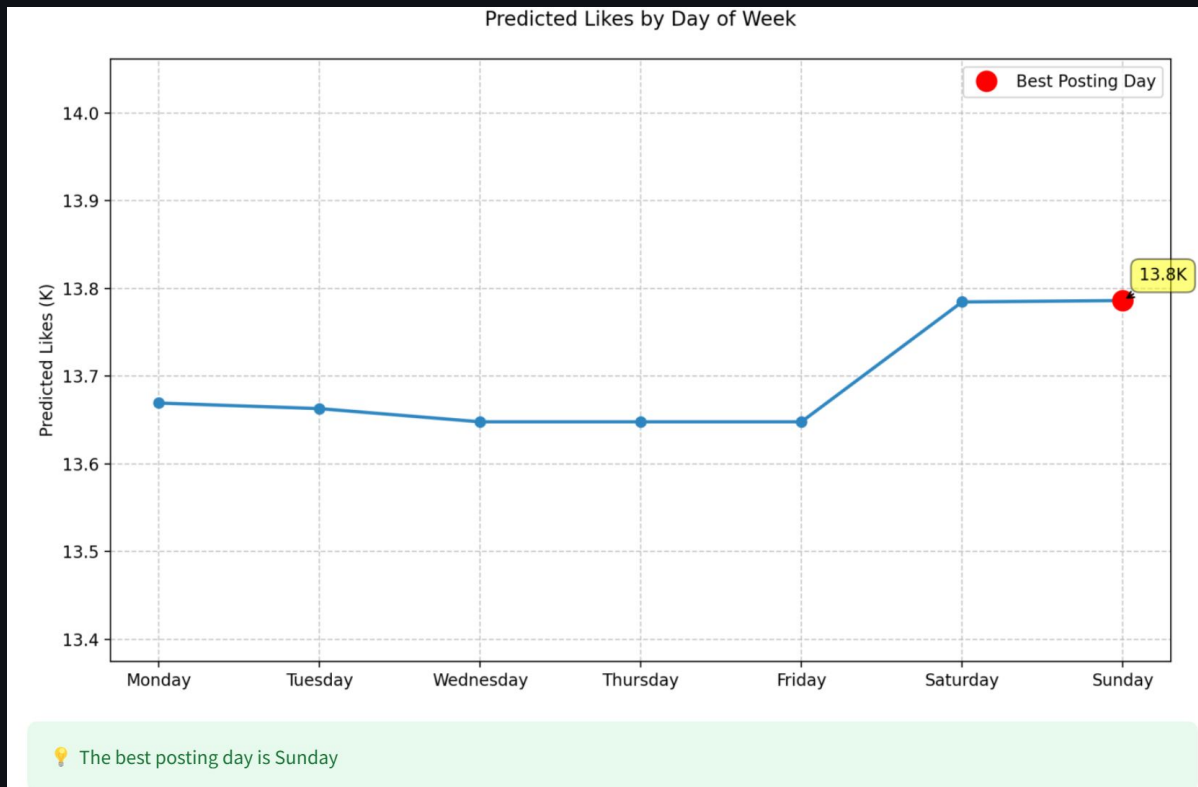
Product Fit: 0.35

Engagement Change: 37.50%

Recommendation 3: movie

- Top Product Picks just for you
- Category & Brand Info so you know what you're promoting
- Predicted Likes — how much your audience might engage
- Scores to help you pick what fits your vibe best
- Engagement Boost % — how much this could grow your stats
- Quick Button to see best posting days for max likes

Tool Elements — Output section cont.



Tool Function

- Predicts how many **likes you'll get** each day of the week
- Highlights the **best posting day** using engagement data
- Gives a **clear visual** to help plan your content timing

Benefits for You

- **Boost your engagement** by posting on high-performing days
- **Save time** — no guessing when to post
- **Improve product promo results** by posting at the right moment
- **Target your audience better** with smart scheduling

Tool Elements — Output section (Hot products only)



Recommended Products

Recommendation 1: movie

Category: Movies_and_TV

Brand: na

Times Promoted: 798

Recommendation 2: album

Category: Digital_Music

Brand: na

Times Promoted: 142

Recommendation 3: book

Category: Books

Brand: na

Times Promoted: 137

Extension

Extra information on how many times the ‘hot’ product being promoted on Instagram

Benefits for You

- **Pick what's trending** so you stay relevant
- **Boost your earnings** by promoting proven products
- **Save time** — no need to search manually
- **Inspire content ideas** based on popular items
- **Build trust** by promoting what others already love

Real-World Application & Ethical Consideration

Real-World Applications

- Help influencers choose products that match their audience and boost engagement
- Assist brands in finding authentic influencer partnerships
- Increase affiliate sales and commission earnings through data-driven promotion
- Optimise content strategy by recommending high-performing product categories
- Save time by automating product discovery and selection

Ethical Considerations

- Transparency: Clearly communicate that product suggestions are algorithm-driven
- Bias Risk: Ensure recommendations aren't unfairly favouring certain brands or categories
- Audience Trust: Encourage influencers to promote only products they truly support
- Data Privacy: Protect influencer and audience data from misuse
- Diversity & Inclusion: Promote a variety of products from different creators and sellers

Limitations

- Limited Product Information: Product names are shown without historical performance data
- Heavy Dependence on Historical Data: the model performs best with influencers who have rich posting history. Cold-start influencers (new or low activity) are harder to recommend for
- Long Runtime for Influencer-Product matching algorithm: The FAISS-powered ensemble model, which combines engagement, fit, and similarity, is effective but computationally intensive
- No Feedback Loop or Manual Overrides: The system doesn't support influencer or brand feedback to refine recommendations

Thank you